

Accommodation & Usage Test Standard Training Material

Based on VF_PTSP_TCOT113

Date: 2025-07-24



What is Accommodation & Usage (A&U)?



Definition:

Accommodation & Usage (A&U) encompasses the entire occupant and user experience within a vehicle's cabin.

Core Focus:

How well the vehicle physically fits its users.

How easily users can access and operate all intended functions.

Impact:

A&U directly influences comfort, user satisfaction, and market success. It is foundational to positive, day-to-day vehicle experiences, making it one of the most important attributes considered in modern automotive design.

1. Accommodation

Focuses on the physical space, layout, and ergonomic suitability of the vehicle for its occupants.

- Ensures that occupants of varying body sizes and shapes can sit comfortably.
- Prioritizes optimal posture, sufficient space, and ease of movement.

2. Usage

Refers to the practical experience of interacting with the vehicle's features and controls.

- Covers accessibility, intuitiveness, and ease of use for all functions.
- Ensures vehicle operation feels natural and requires minimal effort or learning.

Objectives of Accommodation & Usage



- **Ensure Ergonomics Comfort Meets/Exceeds Customer Expectations**
 - Guarantee comfort for all occupants (driver + passengers) across body sizes and demographics.
 - Optimize posture, support, visibility, and usability for short and long journeys.
- **Align Development with Brand Values**
 - Reinforce brand identity through A&U: luxury → refinement, mass-market → robustness & practicality.
 - The cabin experience should embody and communicate the brand's values, ensuring occupants immediately feel the intended identity of the brand.
- **Benchmark Against Competitors**
 - Regularly evaluate competitor vehicles to identify strengths and weaknesses.
 - Use both **quantitative** (dimensions, clearances, visibility) and **qualitative** (perceived comfort, ease of use) benchmarks.
- **Drive Continuous Improvement**
 - Keep evolving ergonomics and usability as customer expectations change.
 - Improve control placement, storage accessibility, and HMI integration.
- **Compliance with Regulatory & Internal Standards**
 - Meet global requirements (FMVSS, ECE, AIS, QCVN) for visibility, accessibility, and safety.
 - Follow internal design standards for consistency across product lines.
- **Enhance User Satisfaction & Market Appeal**
 - Turn ergonomics and usability into a competitive advantage.
 - Boost brand loyalty and positive reviews through superior cabin experience.
- **Support Functional Adaptability**
 - Enable multipurpose use: foldable seating, flexible cargo, practical storage.
 - Ensure accessibility for diverse users, including elderly and limited mobility occupants.

1. Occupant Posture – Comfort & Support, Adjustability and Fatigue Reduction
2. Visibility – All around vision, Blind spot minimization and IPC Visibility
3. Roominess – Leg, Head, Knee, Shoulder, Hip, Arm and Spatial perceived roominess.
4. Ingress/Egress – Door size, Opening angle, Step-in height and clearance
5. Usability & Functionality – Intuitiveness, Responsiveness, user feedback and Seamless functionality
6. Reachability & Accessibility – Control location, ease of use and accessibility
7. Cargo & Storage – Access & Usability, Capacity & Flexibility and Organization

What is Objective Evaluation?



Objective evaluation in ergonomics and usability involves the quantitative measurement of a vehicle's interior dimensions and features, with a focus on factors that directly impact passenger comfort and usability. The assessment covers the following areas:

Roominess: Quantified using standardized dimensions (e.g., SAE J1100, SAE J826) to evaluate headroom, legroom, shoulder room, and knee clearance across all seating positions.

Ingress/Egress: Measured through parameters such as door opening angle, sill height, step-over distance, and ease of access to ensure convenient entry and exit.

Cargo & Storage: Evaluated by measuring luggage volume, storage compartment dimensions, opening size, and accessibility of in-cabin stowage solutions.

Visibility: Assessed using sightline analysis, vision obstruction angles, and mirror field-of-view measurements to confirm safe and comfortable driving conditions.

Occupant Posture: Analyzed with H-point manikins, seating reference points, and body angle measurements (e.g., hip angle, thigh angle, back angle) to confirm natural, comfortable, and supportive seating positions for different user percentiles.

Reachability & Accessibility: Checked using H-point manikins and 3D coordinate tools to verify that controls, displays, and storage areas are within comfortable reach for a range of user percentiles.

Key Characteristics of Objective Evaluation:

Quantitative: Provides precise, numerical measurements rather than subjective impressions.

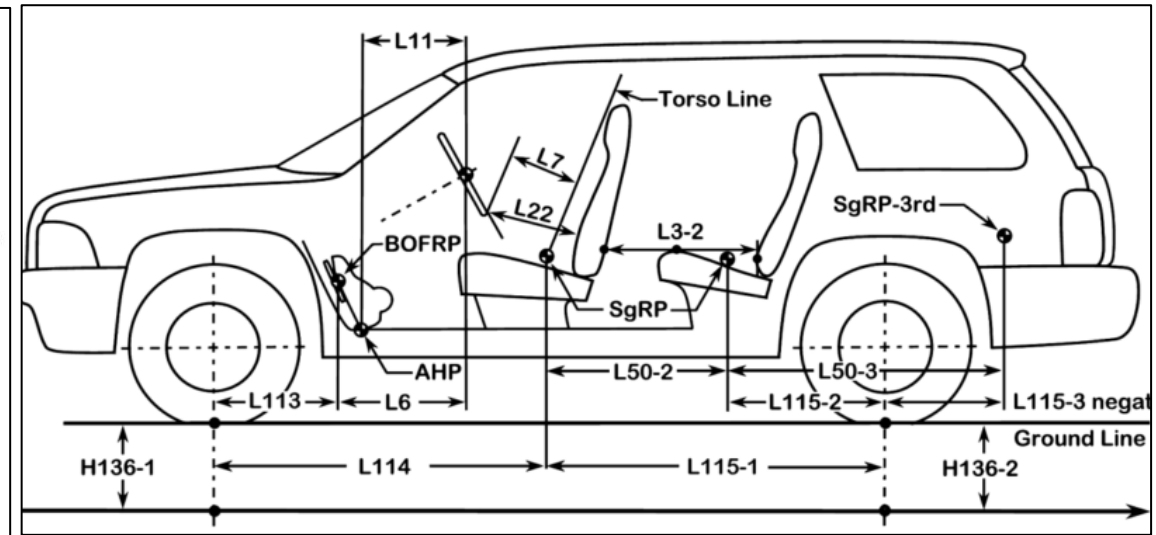
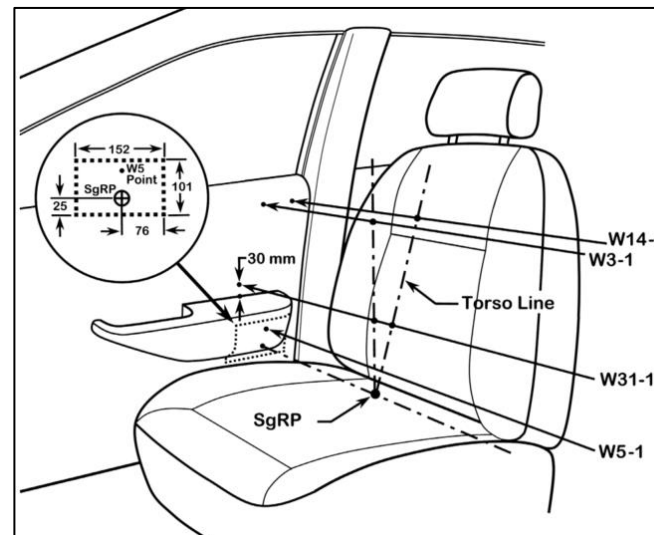
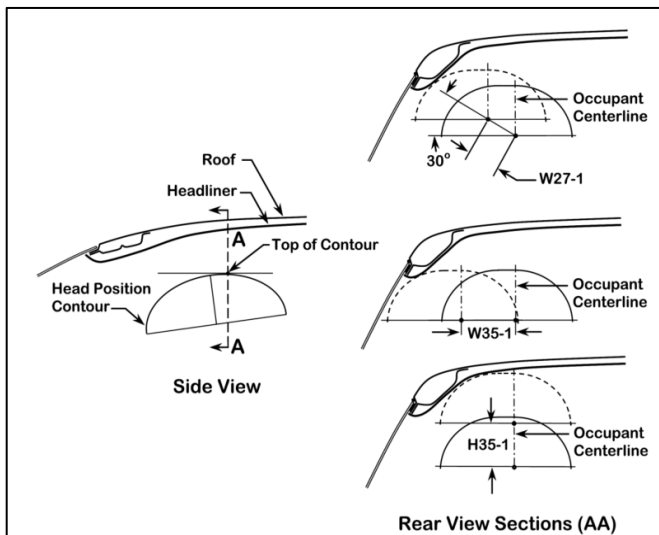
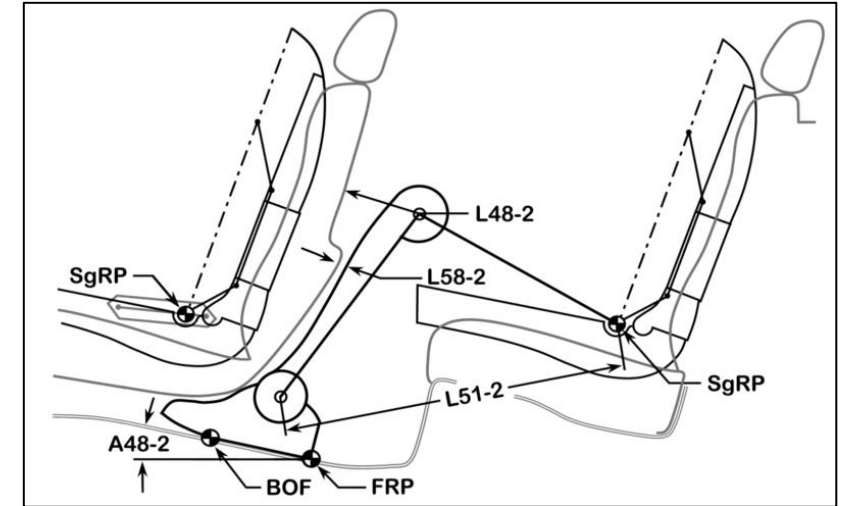
Standardized Methods: Conducted in accordance with global standards such as SAE J1100 and SAE J826.

Equipment-Based: Relies on tools like H-point manikins, 3D scanners, coordinate measuring machines, and measurement rigs.

Repeatable & Reproducible: Ensures results are consistent across different vehicles, enabling benchmarking and a clear understanding of the vehicle's current performance.

Objective Evaluation of Roominess

- **Headroom:** Vertical & Diagonal distance between seat cushion reference and roof/headliner.
- **Legroom:** Lateral & Longitudinal distance between legs and seat/footwell ahead.
- **Knee Room:** Distance between the seat in front and seated occupant's knees.
- **Shoulder Room:** Lateral space at shoulder height.
- **Hip Room:** Space at hip level between door trim and centerline/console.
- **Elbow Room:** Clearance between armrest/door and torso.
- **Footwell Space:** Free space for foot placement, including shoe size allowance.



Objective Evaluation of Ingress/Egress

- **Door Opening Angle:** Degree of door swing, affecting ease of leg and body movement.
- **Step-In Height:** Vertical distance from ground to floor surface.
- **Sill Width:** Horizontal clearance at sill that must be stepped over.
- **Roof Cut-Out / Head Clearance Path:** Clearance between occupant's head trajectory and roof during entry/exit.
- **Knee Clearance:** Free space ahead of knees during leg swing.
- **Seat-to-Door Opening Dimension:** Horizontal distance between seat edge and door inner trim. Large gaps force extra stretch; small gaps cause obstruction.

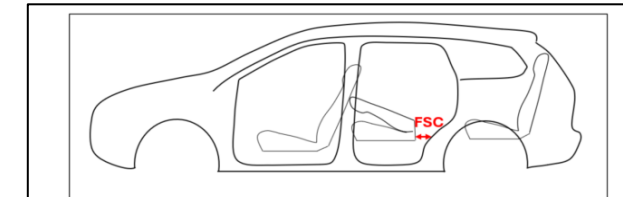
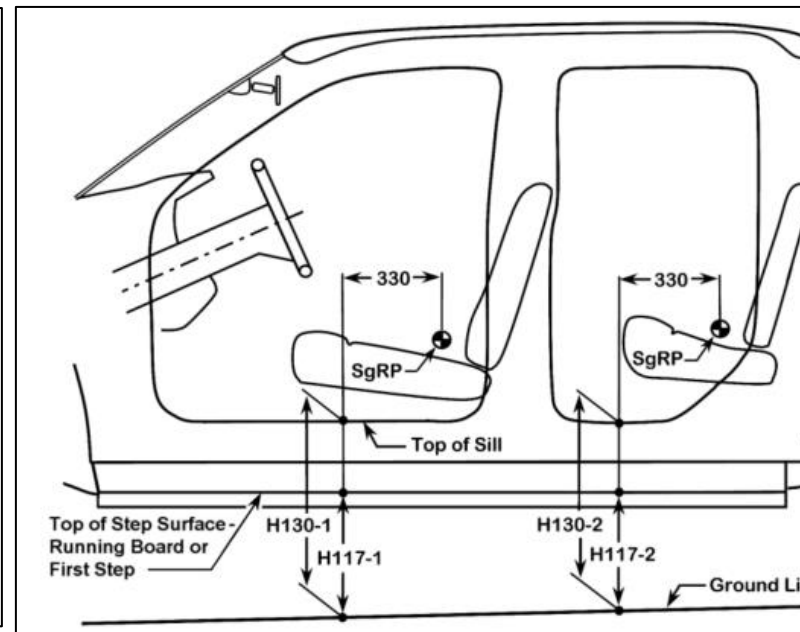
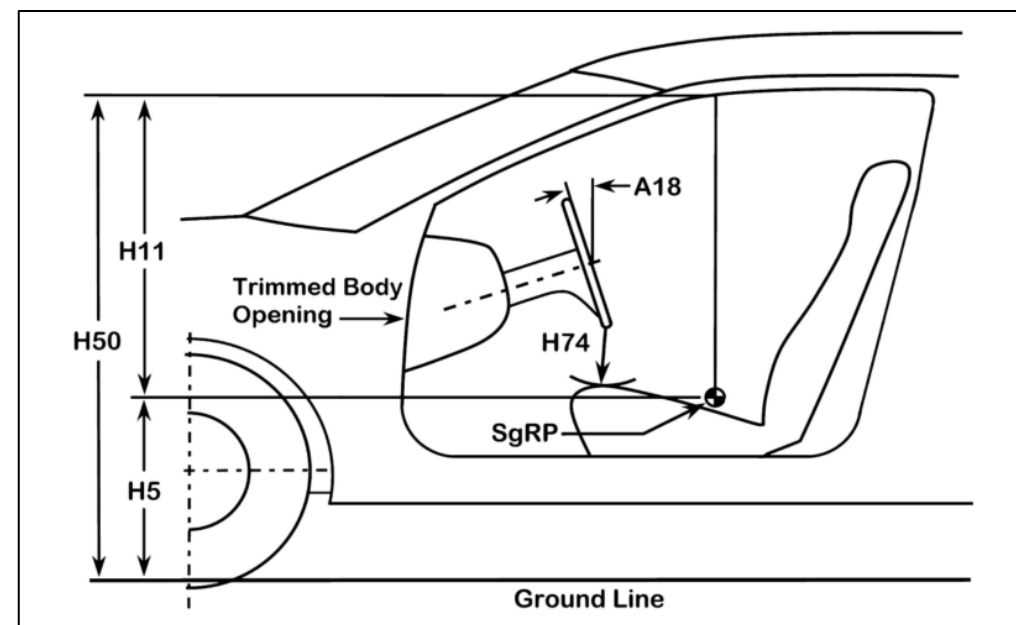
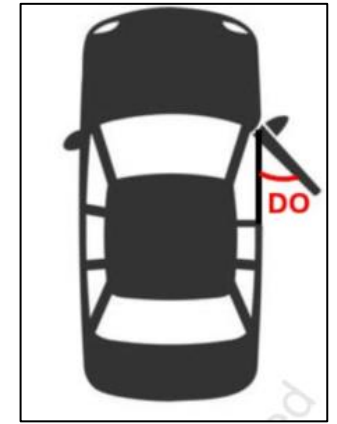
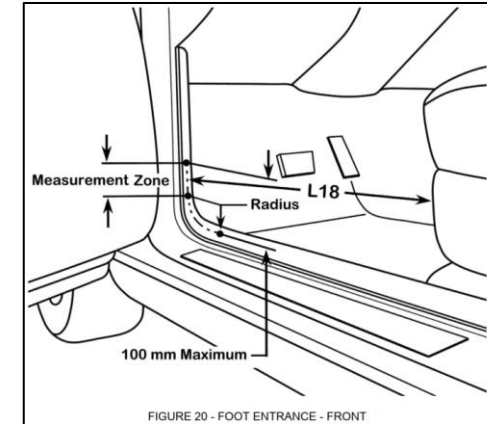


Figure 16

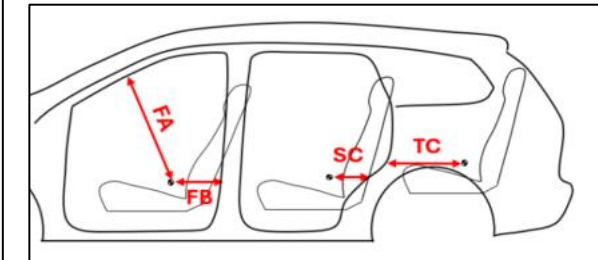


Figure 17

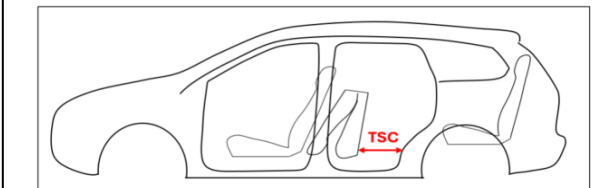
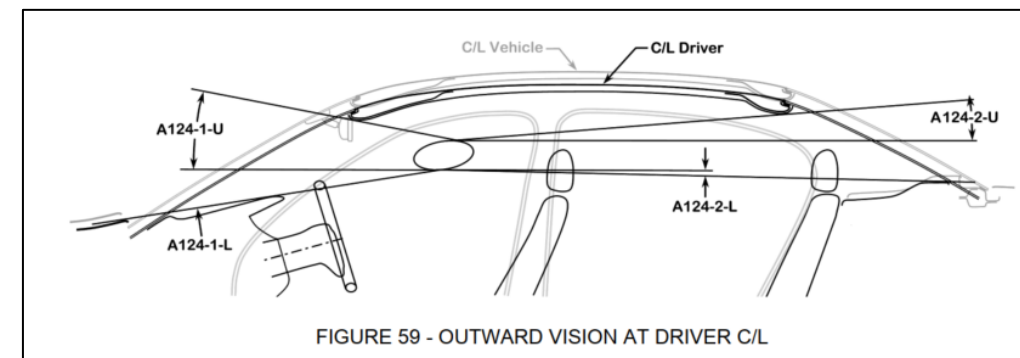
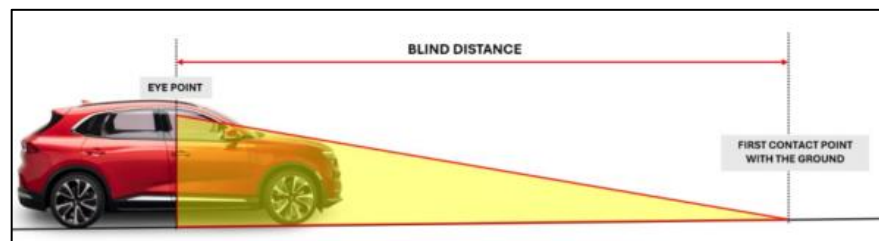
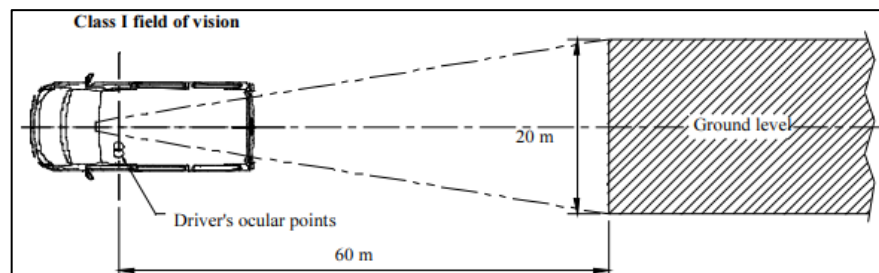
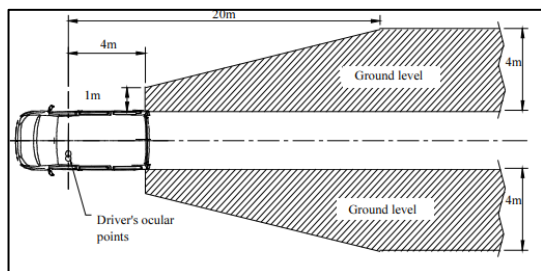


Figure 18

Objective Evaluation of Visibility

- **Forward Field of View:** Minimum visible road length over hood and the 3D visibility.
- **Vision Obscuration by Pillars:**
 - A-Pillar: Critical for pedestrians/traffic lights.
 - B-Pillar: Impacts over-shoulder checks.
 - C-Pillar: Affects reversing and lane changes.
 - D-Pillar: Relevant for SUVs/wagons, affects rear corner vision.
- **Rearward Visibility:** Clear view through rear glass.
- **Mirror Coverage:** Verified against regulatory targets.
- **Instrument Visibility:** Gauges/displays visible without posture change.
- **360° Surround Visibility (through windows):** Continuous vision coverage from left A-pillar, across windshield, side windows, and rear glass, minimizing blind zones



Objective Evaluation of Occupant Posture

Postural Dimensions:

Seat Cushion Angle (Thigh Support): Inclination of seat base relative to horizontal; too steep = pressure on thighs, too flat = insufficient support.

Backrest Angle: Recline angle relative to vertical; must balance visibility and comfort.

Seat

Seat Geometry & Adjustability Dimensions:

Seat Cushion Length (Seat Pan Depth): Distance from front edge of cushion to backrest. Too long = pressure behind knees; too short = poor thigh support.

Seat Cushion Width: Usable lateral width between bolsters. Narrow cushions restrict larger occupants; wide cushions may reduce lateral support.

Seat Backrest Height: Vertical dimension from seat cushion to top of backrest. Determines shoulder and upper back support.

Seat Fore-Aft Travel: Sliding distance to accommodate short and tall occupants.

Backrest Recline Range: Degree of adjustment to allow relaxed posture or upright driving.

Seat Height Adjustment: Vertical movement to suit drivers of different statures.

Steering Wheel Reach & Tilt: Range of telescopic and angular movement for ergonomic arm and shoulder posture.

Accommodation & Usage Attributes

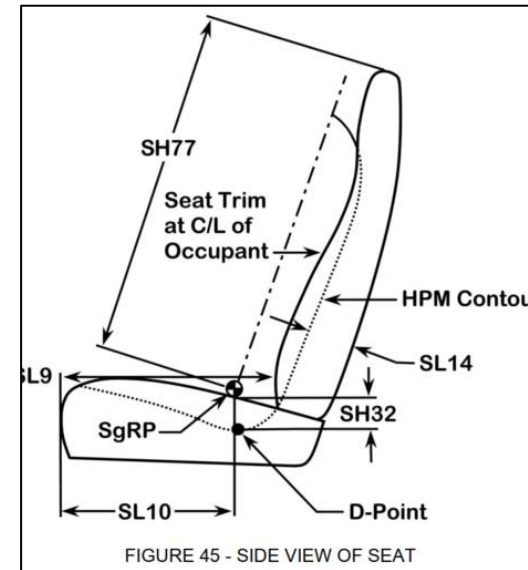


FIGURE 45 - SIDE VIEW OF SEAT

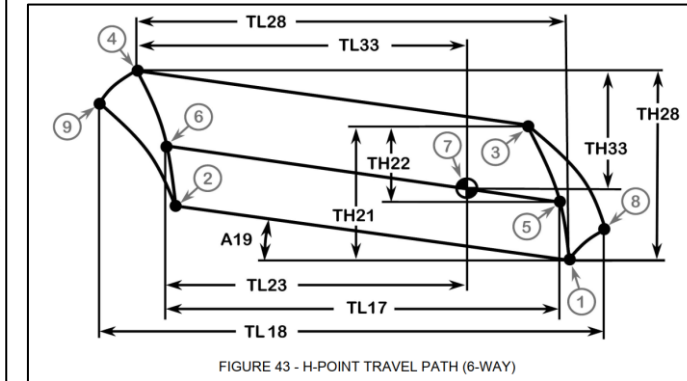


FIGURE 43 - H-POINT TRAVEL PATH (6-WAY)

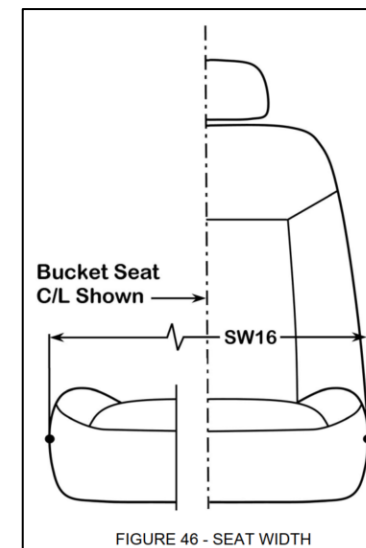


FIGURE 46 - SEAT WIDTH

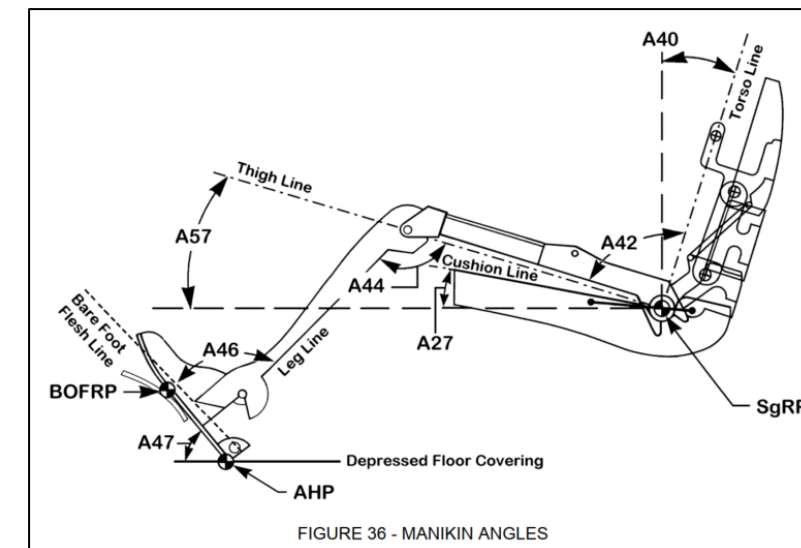


FIGURE 36 - MANIKIN ANGLES

1. Vehicle Preparation

- Confirm the test vehicle is fully assembled, with all features/components present and functional.
- Sign-off by Engineering/Manufacturing/Prototype team, with completed gap and flush check.
- Use a high-spec vehicle if possible; check tire specs, ride height, and overall cleanliness.

2. Equipment Setup

- Calibrate all measuring equipment (e.g., 3D photogrammetry scanner, H-point machine, calibration plate, CMM).
- Set up scanning area: flat, well-lit, minimal reflections or shadows.

3. Test Scenarios

- Perform scans/measures in various scenarios:
 - Exterior (unladen/laden) and underbody scans
 - All closures open (doors, bonnets, trunk)
 - Wiper area scans
 - Interior (with and without seats)
 - Seat positions (standard and extremes)
 - Movable parts at their operational extremes

4. Data Collection

- Place markers/targets as required and position manikins in designated seats for anthropometric measurements.
- Conduct objective measurements according to standards (e.g., SAE J1100, J826, UN ECE R125):
 - Wheelbase, legroom, headroom, shoulder room, entrance/step height, visibility angles, cargo dimensions, etc.
- Document all measurements in the approved format.

5. Data Processing

- Align all scan data using vehicle coordinate references (typically front LH and rear RH wheel centers).
- Process and verify measurements, filling out required DVP (Design Verification Plan) reporting formats.

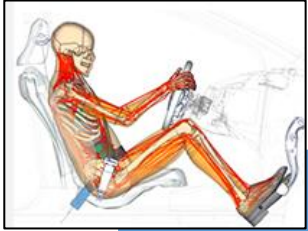
What is Subjective Evaluation?



Subjective evaluation entails a trained assessor's qualitative judgment of the vehicle's comfort, usability, and overall human experience. It focuses on how real users feel and interact with the vehicle's interior and features.

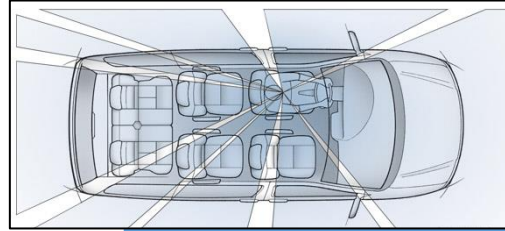
- **Qualitative Focus:** Relies on trained assessors' judgment of vehicle comfort, usability, and human experience based on perception and feedback.
- **Evaluator-Driven:** Conducted systematically by trained individuals across all seating positions and feature interactions.
- **Experience-Oriented:** Judges how intuitive, comfortable, and user-friendly the interior and features feel in real use.
- **Structured Scoring:** Results are captured using a defined scale (e.g., *Best, Good, Acceptable, Poor*), ensuring consistency across assessments.

Subjective Evaluation Parameters



OCCUPANT POSTURE

- Seating Position
- Leg and Foot Position
- Arm and Hand Position
- Head and Neck Alignment
- Pelvic Support
- Shoulder position



VISIBILITY

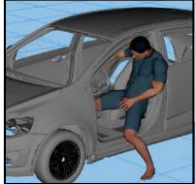
- Forward Visibility
- Peripheral Visibility
- Rear Visibility
- Blind Spot Visibility
- Instrument Panel Visibility
- Headlight and Signal Visibility



ROOMINESS

- Legroom
- Headroom
- Shoulder Room
- Elbow Room
- Hip Room
- Overall Spatial Layout
- Perceived Visibility and Light

Subjective Evaluation Parameters



INGRESS/EGRESS

- Door Opening Size
- Door Swing Range
- Seat Height
- Step-In Height
- Grab Handles and Support
- Seat Adjustability
- Clearance from Obstacles



REACHABILITY & ACCESIBILITY

- Control Location
- Steering Wheel
- Pedals
- Seat
- Storage Compartments
- Ease of Use
- Buttons and Knobs
- Infotainment Systems
- Assistive Technologies



USABILITY

- Intuitive Control Layout
- Ease of Operation
- Clear Instructions
- Accessibility for All Users
- Minimized Cognitive Load
- Reliability
- Versatility
- Efficiency
- User Feedback
- Integration



CARGO & STORAGE

- Accessibility
- Usability
- Capacity
- Flexibility
- Organization
- Security
- Comfort
- Innovative Features

OBJECTIVE

- Provide a natural, comfortable seating position that minimizes fatigue during short and long journeys.
- Ensure posture allows easy control of vehicle functions without strain.
- Support the spine, thighs, and head in a way that feels balanced for all body sizes (5th percentile female to 95th percentile male).
- Allow adjustment so different users can find a comfortable driving or riding position.

EVALUATION SETUP:

- Seat a range of people (short, tall, slim, large build) in the vehicle.
- Ask them to adjust the seat to their preferred driving or riding position.
- Evaluate both driver and passenger seats.

QUESTIONNAIRE EXAMPLE:

•Seat Support:

- Does the seat cushion support the thighs without leaving gaps?
- Is there any sharp edge or hard spot under thighs or hips?

•Backrest:

- Does it follow the natural curve of the spine (lumbar and upper back)?
- Is the lumbar support too strong, too weak, or well-balanced?

•Headrest:

- Does it touch the back of the head naturally or push it forward?
- Is it adjustable enough for both tall and short users?

•Driving Posture:

- Are shoulders relaxed when holding the steering wheel?
- Are knees comfortably bent without being too high or too low?

OBJECTIVE

- Ensure drivers can see the road, surroundings, and key vehicle information without needing to lean or change posture.
- Minimize blind spots caused by pillars, mirrors, or headrests.
- Provide clear forward, side, and rear views under various driving conditions (day/night, rain).
- Keep all critical displays, warning lights, and indicators within an unobstructed line of sight.

EVALUATION SETUP:

Sit in the driver's seat with the seat adjusted for normal driving position.

Perform evaluation in different environments: open road, parking, junctions.

QUESTIONNAIRE EXAMPLE:

•Forward View:

- Can you see the road edge ahead of you without leaning?
- Is the traffic light visible without crouching or moving head excessively?

•Side View:

- Are there significant blind spots caused by A-pillars?
- Can you see pedestrians crossing from the side?

•Rear View:

- Is the rear window large enough to reverse without camera aid?
- Do rear headrests block the view?

•Instrument Visibility:

- Can you read the speedometer and key gauges without changing posture?
- Is there glare on the display from sunlight?

OBJECTIVE

- Provide enough physical and perceived space for occupants to move comfortably in their seating position.
- Prevent contact with the roof, other passengers, or interior panels during normal movement.
- Ensure headroom, legroom, shoulder, and hip space are sufficient for intended passenger capacity.
- Avoid a cramped feeling, even with a full load of passengers.

EVALUATION SETUP:

- Seat multiple people of different sizes in each row.
- Evaluate both driver and passenger space.

QUESTIONNAIRE EXAMPLE:

•Headroom:

- Is there space above the head in a natural seating posture?
- Do taller occupants' heads touch the roof or roof liner?

•Legroom:

- Are knees free from contact with the seat or dashboard ahead?
- Is there enough space to stretch slightly during long rides?

•Shoulder & Hip Space:

- Can two adults sit side-by-side without leaning away?
- Are door panels or center console pressing against hips?

•Foot Space:

- Is there enough space for feet under the front seat?
- Are there any obstructions like seat rails or bulges?

OBJECTIVE

- Allow easy and natural entry and exit for occupants of all sizes, including elderly passengers and those with reduced mobility.
- Avoid awkward movements, bending, or twisting during entry/exit.
- Ensure openings are wide enough, step-in height is comfortable, and roof clearance is sufficient.
- Minimize any contact between occupant body parts and vehicle structures when entering or leaving.

EVALUATION SETUP:

- Observe people entering and exiting the vehicle multiple times.
- Include testers wearing bulky clothes or carrying small bags.

QUESTIONNAIRE EXAMPLE:

•Door Opening:

- Is the opening wide enough to step in without twisting?
- Does the door open far enough for comfortable access?

•Step-In Height:

- Is the floor level easy to step onto without climbing or dropping?

•Head Clearance:

- Do you need to duck to avoid hitting the roof edge?

•Obstructions:

- Do knees or feet hit the door sill, dashboard, or seat base when entering/exiting?
- Are handles easy to grip and positioned well?

OBJECTIVE

- Ensure all features and controls can be located, understood, and operated easily without instruction.
- Make control operation comfortable and quick, with minimal distraction from driving.
- Provide smooth, intuitive interaction with seats, storage compartments, infotainment, climate control, and other features.
- Avoid any excessive force, confusing layouts, or awkward movements during operation.

EVALUATION SETUP:

- Ask testers to operate controls without instruction.
- Include climate controls, infotainment, seat adjustments, storage lids.

QUESTIONNAIRE EXAMPLE:

•Control Location:

- Are common controls (AC, infotainment, lights) easy to locate without looking away from the road?

•Control Effort:

- Do knobs turn smoothly and buttons press without excessive force?

•Intuitiveness:

- Is it clear what each control does without explanation?

•Feedback:

- Do buttons, knobs, and levers give clear tactile or audible feedback when used?

•Storage Access:

- Are compartments easy to open and close with one hand?
- Do lids and covers stay open or close smoothly?

OBJECTIVE

- Ensure all essential controls, storage, and safety items can be reached from the seated position without leaning or stretching uncomfortably.
- Provide logical placement so frequently used items are closest to hand.
- Ensure accessibility for both tall and short users, including rear seat passengers.
- Make sure seat belts, handles, and emergency controls are always easy to access.

EVALUATION SETUP:

- Seat testers in normal position and ask them to reach for specific items (seat belt, cup holder, glove box, infotainment screen).

QUESTIONNAIRE EXAMPLE:

•Primary Controls:

- Can all occupants reach seat belts, gear shifter, and main controls without leaning forward excessively?

•Storage Reach:

- Can bottles, glovebox, and center console storage be reached from seated position comfortably?

•Emergency Items:

- Can hazard light button, parking brake, and window controls be accessed instantly?

•Passenger Access:

- Can rear passengers easily reach charging ports, seat pockets, and cup holders?

OBJECTIVE

- Provide adequate and practical space for luggage, groceries, and other personal items.
- Ensure loading/unloading is easy, with an opening and height suited to the intended use.
- Keep storage compartments secure, accessible, and sized for real-world items.
- Offer flexible cargo configurations (e.g., foldable seats) to adapt to different load needs.

EVALUATION SETUP:

- Ask testers to place and remove common items (bottles, bags, suitcases) in storage spaces.
- Include luggage loading and unloading.

QUESTIONNAIRE EXAMPLE:

• Loading Access:

- Is the boot/trunk opening wide and tall enough for typical items?
- Is the loading height comfortable for lifting heavy items?

• Storage Shape:

- Is the cargo area flat and regular in shape, or are there awkward bulges?

• Small Storage:

- Do cup holders grip bottles firmly without rattling?
- Can small items (phone, keys) be placed securely and retrieved easily?

• Seat Folding:

- Can seats fold down easily without heavy lifting or complicated steps?
- Is the cargo floor flat when seats are folded?

Subjective Evaluation Scoring

SCORING CRITERIA

Rating	Score	Detailed Description
Best	>8	Outstanding experience that feels effortless and intuitive. Comfort and usability exceed expectations, with no noticeable flaws. Vehicle interaction feels premium, refined, and highly user-centric. Creates a strong “wow” factor and sets a benchmark in its class.
Good	7–8	Comfortable and easy to use. Well-designed and functional, with smooth operation in most aspects. Minor improvements could elevate it to best-in-class, but no real complaints from users. Meets or slightly exceeds customer expectations consistently.
Acceptable	6–7	Generally positive experience but with some noticeable flaws or areas for improvement. Still usable and comfortable for most users, but certain details may require compromise. Customers may raise small complaints.
Basic	5–6	Meets only the minimum expectations. Usable, but lacks refinement and polish. Noticeable discomfort, awkward controls, or poor adjustability reduce satisfaction. Customers would see it as cost-driven or entry-level rather than user-focused.
Unacceptable	<5	Falls below basic expectations. Significant discomfort, poor usability, or glaring ergonomic issues make it difficult to use. Customers perceive it as a clear weakness and would likely reject the product.

Subjective Scoring –Examples



Best (>8)

“The seat felt perfectly tailored. Thigh, spine, and head support were seamless and natural. Long drive comfort was luxury-class, exceedingly even segment benchmarks.”

Good (7–8)

“Comfortable and natural posture for both tall and short users. Adjustments were easy, and no complaints were noted by participants.”

Acceptable (6–7)

“Posture was mostly fine, but taller drivers felt the steering wheel reach a little short. Some users noted lumbar support could be stronger. Minor complaints but still functional.”

Basic (5–6)

“Seat cushion was short for tall users, causing thigh fatigue. Short drivers struggled with pedal reach. Several complaints across different body sizes.”

Unacceptable (<5)

“Drivers experienced immediate discomfort. Seat height, cushion, and backrest geometry forced awkward posture. Testers said they couldn’t drive more than 20 minutes comfortably.”

Subjective Scoring – Examples



Best (>8)

“Entry and exit felt effortless regardless of height/age. Doors open wide, sill height is low, and movement felt luxury-class — beyond anything in segment.”

Good (7–8)

“Ingress and egress were smooth for all occupants. Door swing and sill height were well designed. Nothing to complain about.”

Acceptable (6–7)

“Generally okay, but taller passengers had to duck slightly when entering. Complaints are minor but noticeable in comparison to competitors.”

Basic (5–6)

“Elderly passengers struggled with the high step-in. Complaints included narrow door angle and need to twist body while entering.”

Unacceptable (<5)

“Ingress was a major pain point. Rear entry required crouching and twisting. Elderly passengers needed physical support to get out.”

Why should we conduct subjective evaluation in addition to objective evaluation?

1. Human Perception Cannot Be Fully Quantified

- Objective evaluation gives numbers (dimensions, angles, pressures, reach distances).
- But comfort, ease of use, and perceived quality are *experienced*, not just measured.
- Example: Two seats may both meet the same seat cushion angle and thigh support requirement, but one “feels” softer and more premium, which only users can judge.

2. Bridges the Gap Between Standards and Real Use

- Subjective testing shows whether meeting those numbers translates into a positive user experience.
- Example: A vehicle may meet the minimum rear visibility requirement, but thick C-pillars might still make drivers feel unsafe.

3. Captures Emotional and Brand Experience

- Customers buy based on *perception of comfort and usability*, not only technical specs.
- Subjective evaluation measures alignment with **brand promise** (premium, sporty, family-friendly).
- Example: A luxury SUV and an economy hatchback may have the same ingress step height, but expectations for “ease of entry” differ greatly.

4. Identifies Nuances Missed by Objective Data

- Objective data can’t capture:
 - “Does this knob feel intuitive to use?”
 - “Do I feel cramped even though headroom is technically sufficient?”
 - “Does the cabin design feel relaxing on long trips?”
- These subtleties often drive customer satisfaction and reviews.

5. Validates Real-World Usability

- Lab-based objective measures might not reflect actual use cases.
- Subjective evaluation in test drives, stop-go traffic, night driving, or long journeys reveals hidden issues.
- Example: Pedal placement might be within spec, but drivers may report ankle strain after 2 hours.

6. Competitive Benchmarking

- Objective specs can be very close between competitors.
- Subjective scoring shows which product feels “better” in customer eyes.
- Example: Two cars may offer nearly identical rear legroom dimensions, but customers often rate one roomier based on seat contour and cushion angle.

END